

Programming patterns – special problems



Special problems

- two return value in a function
- Which data should control the solution/execution
- constructions with copy
- problems with interval
- transformation of input data



Function with 2 (more) return values

Problem: A map is stored in a grid (matrix). At each cell, an integer tells you the high of the point there. Identify the point on the map that is in the furthest northwest and contains a lake.

40	74	42	20	10	51	61	23	6	89	75	46	40	28	27	79	91	33	99	27	51	92
54	44	53	0	70	14	72	72	59	87	66	82	26	66	78	65	55	91	59	77	67	73
5	16	0	0	0	0	24	70	49	90	89	43	92	9	62	3	89	56	21	47	50	25
38	69	35	0	0	0	0	19	88	19	99	1	80	##	27	54	46	77	55	32	46	31
59	43	56	0	0	26	5	86	56	92	49	88	18	67	81	45	85	26	79	74	41	36
57	13	80	69	38	35	32	25	89	43	48	20	62	81	49	94	6	88	47	2	57	91
12	22	21	41	12	52	84	95	40	13	48	36	55	22	50	48	64	52	##	73	98	49
86	57	70	14	80	98	36	45	49	32	28	53	58	99	36	40	93	90	82	74	98	14
57	12	45	36	75	36	4	58	64	54	87	3	0	0	0	57	75	26	44	30	35	43
80	10	65	1	1	19	12	69	38	24	6	62	0	0	22	27	85	73	31	2	87	56
9	89	18	54	37	94	78	80	16	21	93	96	85	0	8	81	30	87	1	75	80	40
##	83	70	63	2	29	47	29	68	24	4	50	96	92	95	90	29	97	90	86	11	99
10	4	60	64	31	55	72	84	69	12	43	28	35	74	18	11	54	64	73	48	60	76

1st version

Problem: In a matrix of integers, which is the row and column where 0 is the first number in the sequence?

Another way: In a matrix of integers, which is the first row in which 0 occurs, and where does it occur first?

=> Decision and Selection in a Search

40	74	42	20	10	51	61	23	6	89	75	46	40	28	27	79	91	33	99	27	51	92
54	44	53	0	70	14	72	72	59	87	66	82	26	66	78	65	55	91	59	77	67	73
5	16	0	0	0	0	24	70	49	90	89	43	92	9	62	3	89	56	21	47	50	25
38	69	35	0	0	0	0	19	88	19	99	1	80	##	27	54	46	77	55	32	46	31
59	43	56	0	0	26	5	86	56	92	49	88	18	67	81	45	85	26	79	74	41	36
57	13	80	69	38	35	32	25	89	43	48	20	62	81	49	94	6	88	47	2	57	91
12	22	21	41	12	52	84	95	40	13	48	36	55	22	50	48	64	52	##	73	98	49
86	57	70	14	80	98	36	45	49	32	28	53	58	99	36	40	93	90	82	74	98	14
57	12	45	36	75	36	4	58	64	54	87	3	0	0	0	57	75	26	44	30	35	43
80	10	65	1	1	19	12	69	38	24	6	62	0	0	22	27	85	73	31	2	87	56
9	89	18	54	37	94	78	80	16	21	93	96	85	0	8	81	30	87	1	75	80	40
##	83	70	63	2	29	47	29	68	24	4	50	96	92	95	90	29	97	90	86	11	99
10	4	60	64	31	55	72	84	69	12	43	28	35	74	18	11	54	64	73	48	60	76



1st version

40	74	42	20	10	51	61	23	6	89	75	46	40	28	27	79	91	33	99	27	51	92
54	44	53	0	70	14	72	72	59	87	66	82	26	66	78	65	55	91	59	77	67	73
5	16	0	0	0	0	24	70	49	90	89	43	92	9	62	3	89	56	21	47	50	25
38	69	35	0	0	0	0	19	88	19	99	1	80	##	27	54	46	77	55	32	46	31
59	43	56	0	0	26	5	86	56	92	49	88	18	67	81	45	85	26	79	74	41	36
57	13	80	69	38	35	32	25	89	43	48	20	62	81	49	94	6	88	47	2	57	91
12	22	21	41	12	52	84	95	40	13	48	36	55	22	50	48	64	52	##	73	98	49
86	57	70	14	80	98	36	45	49	32	28	53	58	99	36	40	93	90	82	74	98	14
57	12	45	36	75	36	4	58	64	54	87	3	0	0	0	57	75	26	44	30	35	43
80	10	65	1	1	19	12	69	38	24	6	62	0	0	22	27	85	73	31	2	87	56
9	89	18	54	37	94	78	80	16	21	93	96	85	0	8	81	30	87	1	75	80	40
##	83	70	63	2	29	47	29	68	24	4	50	96	92	95	90	29	97	90	86	11	99
10	4	60	64	31	55	72	84	69	12	43	28	35	74	18	11	54	64	73	48	60	76

Problem:

In a matrix of integers, which is the first row in which 0 occurs, and where does it occur first?

Specification:

In: $n \in \mathbb{N}$, $m \in \mathbb{N}$, $\text{matrix} \in \mathbb{Z}[1..n, 1..m]$

Out: $\text{van} \in \mathbb{L}$, $\text{sind} \in \mathbb{N}$, $\text{oind} \in \mathbb{N}$

Fn: $\text{exnull}: \mathbb{N} \rightarrow \mathbb{L}$, $\text{exnull}(i) = \text{DECISION}(j=1..m, \text{matrix}[i,j]=0)$

Pre: -

Post: $(\text{exists}, \text{sind}) = \text{SEARCH}(i=1..n, \text{exnull}(i))$ and
 $\text{exists} \rightarrow \text{oind} = \text{SELECT}(j \geq 1, \text{matrix}[\text{sind}, j]=0)$



1st version

40	74	42	20	10	51	61	23	6	89	75	46	40	28	27	79	91	33	99	27	51	92
54	44	53	0	70	14	72	72	59	87	66	82	26	66	78	65	55	91	59	77	67	73
5	16	0	0	0	0	24	70	49	90	89	43	92	9	62	3	89	56	21	47	50	25
38	69	35	0	0	0	0	19	88	19	99	1	80	##	27	54	46	77	55	32	46	31
59	43	56	0	0	26	5	86	56	92	49	88	18	67	81	45	85	26	79	74	41	36
57	13	80	69	38	35	32	25	89	43	48	20	62	81	49	94	6	88	47	2	57	91
12	22	21	41	12	52	84	95	40	13	48	36	55	22	50	48	64	52	##	73	98	49
86	57	70	14	80	98	36	45	49	32	28	53	58	99	36	40	93	90	82	74	98	14
57	12	45	36	75	36	4	58	64	54	87	3	0	0	0	57	75	26	44	30	35	43
80	10	65	1	1	19	12	69	38	24	6	62	0	0	22	27	85	73	31	2	87	56
9	89	18	54	37	94	78	80	16	21	93	96	85	0	8	81	30	87	1	75	80	40
##	83	70	63	2	29	47	29	68	24	4	50	96	92	95	90	29	97	90	86	11	99
10	4	60	64	31	55	72	84	69	12	43	28	35	74	18	11	54	64	73	48	60	76

Specification:

In: $n \in \mathbb{N}$, $m \in \mathbb{N}$, $\text{matrix} \in \mathbb{Z}[1..n, 1..m]$

Out: $\text{van} \in L$, $\text{sind} \in \mathbb{N}$, $\text{oind} \in \mathbb{N}$

Fn: $\text{exnull}: \mathbb{N} \rightarrow L$, $\text{exnull}(i) = \text{DECISION}(j=1..m, \text{matrix}[i,j]=0)$

Pre: -

Post: $(\text{exists}, \text{sind}) = \text{SEARCH}(i=1..n, \text{exnull}(i))$ and
 $\text{exists} \rightarrow \text{oind} = \text{SELECT}(j \geq 1, \text{matrix}[\text{sind}, j]=0)$

Search

$\text{ind} \sim \text{sind}$

$\text{b..e} \sim 1..n$

$A(i) \sim \text{vannulla}(i)$

Decision (exnull)

$i \sim j$

$\text{b..e} \sim 1..m$

$A(i) \sim \text{matrix}[i,j]=0$

Select

$\text{ind} \sim \text{oind}$

$i \sim j$

$b \sim 1$

$A(i) \sim \text{matrix}[\text{sind}, j]=0$



1st version

40	74	42	20	10	51	61	23	6	89	75	46	40	28	27	79	91	33	99	27	51	92
54	44	53	0	70	14	72	72	59	87	66	82	26	66	78	65	55	91	59	77	67	73
5	16	0	0	0	0	24	70	49	90	89	43	92	9	62	3	89	56	21	47	50	25
38	69	35	0	0	0	0	19	88	19	99	1	80	##	27	54	46	77	55	32	46	31
59	43	56	0	0	26	5	86	56	92	49	88	18	67	81	45	85	26	79	74	41	36
57	13	80	69	38	35	32	25	89	43	48	20	62	81	49	94	6	88	47	2	57	91
12	22	21	41	12	52	84	95	40	13	48	36	55	22	50	48	64	52	##	73	98	49
86	57	70	14	80	98	36	45	49	32	28	53	58	99	36	40	93	90	82	74	98	14
57	12	45	36	75	36	4	58	64	54	87	3	0	0	0	57	75	26	44	30	35	43
80	10	65	1	1	19	12	69	38	24	6	62	0	0	22	27	85	73	31	2	87	56
9	89	18	54	37	94	78	80	16	21	93	96	85	0	8	81	30	87	1	75	80	40
##	83	70	63	2	29	47	29	68	24	4	50	96	92	95	90	29	97	90	86	11	99
10	4	60	64	31	55	72	84	69	12	43	28	35	74	18	11	54	64	73	48	60	76

Post: $(\text{exists}, \text{sind}) = \text{SEARCH}(i=1..n, \text{exnull}(i))$ and
 $\text{exists} \rightarrow \text{oind} = \text{SELECT}(j \geq 1, \text{matrix}[\text{sind}, j] = 0)$

Search

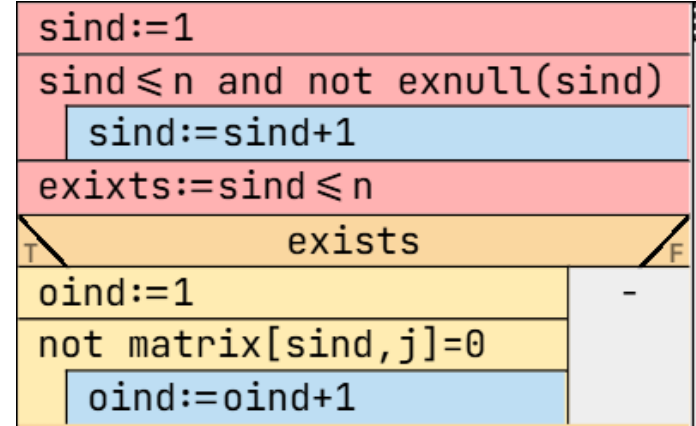
$\text{ind} \sim \text{sind}$
 $\text{b..e} \sim 1..n$
 $A(i) \sim \text{vannulla}(i)$

Decision (exnull)

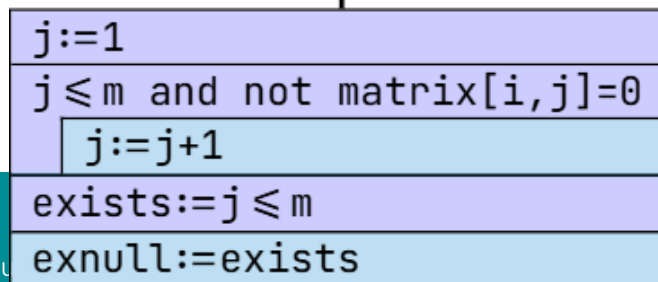
$i \sim j$
 $\text{b..e} \sim 1..m$
 $A(i) \sim \text{matrix}[i, j] = 0$

Select

$\text{ind} \sim \text{oind}$
 $i \sim j$
 $b \sim 1$
 $A(i) \sim \text{matrix}[\text{sind}, j] = 0$



$\text{exnull}(i:\text{Integer}): \text{Boolean}$



2nd version

Problem: In a matrix of integers, which is the row and column where 0 is the first number in the sequence?

Another way: In a matrix of integers, which is the first row in which 0 occurs, and where does it occur first?

=> Search and Search in a Search

40	74	42	20	10	51	61	23	6	89	75	46	40	28	27	79	91	33	99	27	51	92
54	44	53	0	70	14	72	72	59	87	66	82	26	66	78	65	55	91	59	77	67	73
5	16	0	0	0	0	24	70	49	90	89	43	92	9	62	3	89	56	21	47	50	25
38	69	35	0	0	0	0	19	88	19	99	1	80	##	27	54	46	77	55	32	46	31
59	43	56	0	0	26	5	86	56	92	49	88	18	67	81	45	85	26	79	74	41	36
57	13	80	69	38	35	32	25	89	43	48	20	62	81	49	94	6	88	47	2	57	91
12	22	21	41	12	52	84	95	40	13	48	36	55	22	50	48	64	52	##	73	98	49
86	57	70	14	80	98	36	45	49	32	28	53	58	99	36	40	93	90	82	74	98	14
57	12	45	36	75	36	4	58	64	54	87	3	0	0	0	57	75	26	44	30	35	43
80	10	65	1	1	19	12	69	38	24	6	62	0	0	22	27	85	73	31	2	87	56
9	89	18	54	37	94	78	80	16	21	93	96	85	0	8	81	30	87	1	75	80	40
##	83	70	63	2	29	47	29	68	24	4	50	96	92	95	90	29	97	90	86	11	99
10	4	60	64	31	55	72	84	69	12	43	28	35	74	18	11	54	64	73	48	60	76



2nd version

40	74	42	20	10	51	61	23	6	89	75	46	40	28	27	79	91	33	99	27	51	92
54	44	53	0	70	14	72	72	59	87	66	82	26	66	78	65	55	91	59	77	67	73
5	16	0	0	0	0	24	70	49	90	89	43	92	9	62	3	89	56	21	47	50	25
38	69	35	0	0	0	0	19	88	19	99	1	80	##	27	54	46	77	55	32	46	31
59	43	56	0	0	26	5	86	56	92	49	88	18	67	81	45	85	26	79	74	41	36
57	13	80	69	38	35	32	25	89	43	48	20	62	81	49	94	6	88	47	2	57	91
12	22	21	41	12	52	84	95	40	13	48	36	55	22	50	48	64	52	##	73	98	49
86	57	70	14	80	98	36	45	49	32	28	53	58	99	36	40	93	90	82	74	98	14
57	12	45	36	75	36	4	58	64	54	87	3	0	0	0	57	75	26	44	30	35	43
80	10	65	1	1	19	12	69	38	24	6	62	0	0	22	27	85	73	31	2	87	56
9	89	18	54	37	94	78	80	16	21	93	96	85	0	8	81	30	87	1	75	80	40
##	83	70	63	2	29	47	29	68	24	4	50	96	92	95	90	29	97	90	86	11	99
10	4	60	64	31	55	72	84	69	12	43	28	35	74	18	11	54	64	73	48	60	76

Problem:

In a matrix of integers, which is the first row in which 0 occurs, and where does it occur first?

```
searchnull(i)=(exists,ind):((exists,ind)=SEARCH(j=1..m, matrix[i,j]>0))
```

Specification:

In: $n \in \mathbb{N}$, $m \in \mathbb{N}$, $\text{matrix} \in \mathbb{Z}[1..n, 1..m]$

Out: $\text{exists} \in \mathbb{L}$, $\text{send} \in \mathbb{N}$, $\text{oind} \in \mathbb{N}$

Fn: $\text{searchnull}: \mathbb{N} \rightarrow \mathbb{L} \times \mathbb{N}$,

$\text{searchnull}(i) = \text{SEARCH}(j=1..m, \text{matrix}[i,j]=0)$

Pre: -

Post: $(\text{exists}, \text{send}) = \text{SEARCH}(i=1..n, \text{searchnull}(i).\text{exists})$
and $\text{exists} \rightarrow \text{oind} = \text{searchnull}(i).\text{ind}$



2nd version

40	74	42	20	10	51	61	23	6	89	75	46	40	28	27	79	91	33	99	27	51	92
54	44	53	0	70	14	72	72	59	87	66	82	26	66	78	65	55	91	59	77	67	73
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38	69	35	0	0	0	0	19	88	19	99	1	80	##	27	54	46	77	55	32	46	31
59	43	56	0	0	26	5	86	56	92	49	88	18	67	81	45	85	26	79	74	41	36
57	13	80	69	38	35	32	25	89	43	48	20	62	81	49	94	6	88	47	2	57	91
12	22	21	41	12	52	84	95	40	13	48	36	55	22	50	48	64	52	##	73	98	49
86	57	70	14	80	98	36	45	49	32	28	53	58	99	36	40	93	90	82	74	98	14
57	12	45	36	75	36	4	58	64	54	87	3	0	0	0	57	75	26	44	30	35	43
80	10	65	1	1	19	12	69	38	24	6	62	0	0	22	27	85	73	31	2	87	56
9	89	18	54	37	94	78	80	16	21	93	96	85	0	8	81	30	87	1	75	80	40
##	83	70	63	2	29	47	29	68	24	4	50	96	92	95	90	29	97	90	86	11	99
10	4	60	64	31	55	72	84	69	12	43	28	35	74	18	11	54	64	73	48	60	76

Specification:

In: $n \in \mathbb{N}$, $m \in \mathbb{N}$, $\text{matrix} \in \mathbb{Z}[1..n, 1..m]$

Out: $\text{exists} \in \mathbb{L}$, $\text{sind} \in \mathbb{N}$, $\text{oind} \in \mathbb{N}$

Fn: $\text{searchnull}: \mathbb{N} \rightarrow \mathbb{L} \times \mathbb{N}$,

$\text{searchnull}(i) = \text{SEARCH}(j=1..m, \text{matrix}[i,j]=0)$

Pre: -

Post: $(\text{exists}, \text{sind}) = \text{SEARCH}(i=1..n, \text{searchnull}(i).\text{exists})$
and $\text{exists} \rightarrow \text{oind} = \text{searchnull}(i).\text{ind}$

Search

$\text{ind} \sim \text{sind}$

$b..e \sim 1..n$

$A(i) \sim \text{searchnull}(i).\text{exists}$

Search (searchnull)

$i \sim j$

$b..e \sim 1..m$

$A(i) \sim \text{matrix}[i,j]=0$



2nd version

40	74	42	20	10	51	61	23	6	89	75	46	40	28	27	79	91	33	99	27	51	92
54	44	53	0	70	14	72	72	59	87	66	82	26	66	78	65	55	91	59	77	67	73
5	16	0	0	0	0	24	70	49	90	89	43	92	9	62	3	89	56	21	47	50	25
38	69	35	0	0	0	0	19	88	19	99	1	80	##	27	54	46	77	55	32	46	31
59	43	56	0	0	26	5	86	56	92	49	88	18	67	81	45	85	26	79	74	41	36
57	13	80	69	38	35	32	25	89	43	48	20	62	81	49	94	6	88	47	2	57	91
12	22	21	41	12	52	84	95	40	13	48	36	55	22	50	48	64	52	##	73	98	49
86	57	70	14	80	98	36	45	49	32	28	53	58	99	36	40	93	90	82	74	98	14
57	12	45	36	75	36	4	58	64	54	87	3	0	0	0	57	75	26	44	30	35	43
80	10	65	1	1	19	12	69	38	24	6	62	0	0	22	27	85	73	31	2	87	56
9	89	18	54	37	94	78	80	16	21	93	96	85	0	8	81	30	87	1	75	80	40
##	83	70	63	2	29	47	29	68	24	4	50	96	92	95	90	29	97	90	86	11	99
10	4	60	64	31	55	72	84	69	12	43	28	35	74	18	11	54	64	73	48	60	76

Post: $(exists, sind) = \text{SEARCH}(i=1..n, \text{searchnull}(i).exists)$
 and exists $\rightarrow oind = \text{searchnull}(i).ind$

Search

ind $\sim$ sind

b..e $\sim$ 1..n

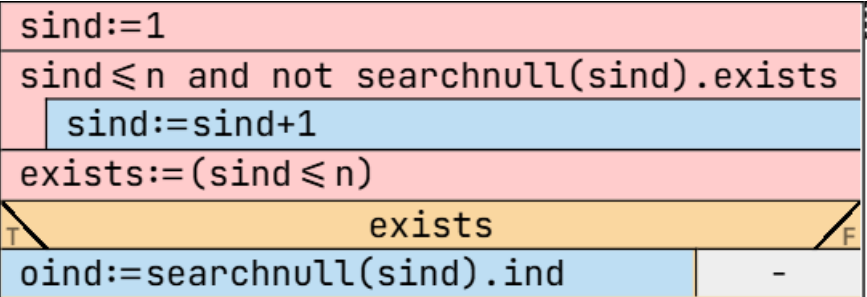
A(i) $\sim$ searchnull(i).exists

Search (searchnull)

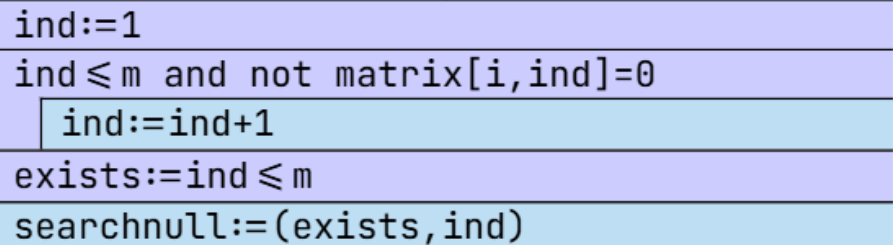
i $\sim$ j

b..e $\sim$ 1..m

A(i) $\sim$ matrix[i,j]=0



searchnull(i:Integer): (Boolean,Integer)



Which data should control the solution/execution

Problem:

There are m guards in the City Watch, and we know which guard was on duty on each of the n days. Which guard was on duty most days?



'Night Watch' - 2002 by Paul Kidby (from Terry Pratchett's Discworld series) – see Rembrandt's Night Watch

1. version: based on the day



Problem:

There are m guards in the City Watch, and we know which guard was on duty on each of the n days. Which guard was on duty **most days**?

Idea:

1. For each **day**, we determine **how many times** the guard has been on duty that day
 2. We choose the **day** where **this number** is **the highest**
- => **counting** in a **maximum selection**

1	3
2	5
3	4
4	5
5	4
6	6
7	5
n=8	3

1. version: day



Problem:

There are m guards in the City Watch, and we know which guard was on duty on each of the n days. Which guard was on duty **most days**?

Specification:

Íŋ $n \in \mathbb{N}$ ~~$n \in \mathbb{N}$~~ $g_{u\grave{a}s\grave{d}} \in \mathbb{N}$, n

Auŋ $i \in \mathbb{N}$

Ôuŋ $d \in \mathbb{N}$

Gŋ $h_n \mathbb{N} \mathbb{N}$ **hŋ $d\grave{a}y$** CÔUŋŋ i $d\grave{a}y$, n $g_{u\grave{a}s\grave{d}}$ i $g_{u\grave{a}s\grave{d}}$ $d\grave{a}y$

Rsê

Rôŋŋ $i \in \mathbb{N}$ $N\grave{A}Y$ **$d\grave{a}y$** , n **hŋ $d\grave{a}y$**
 $\grave{a}ŋ\grave{d}$ $d \in \mathbb{N}$ $g_{u\grave{a}s\grave{d}}$ $i \in \mathbb{N}$ $\grave{a}ŋ\grave{d}$ ~~n~~

1	3
2	5
3	4
4	5
5	4
6	6
7	5
n=8	3

1. version: day

Algorithm:



1	3
2	5
3	4
4	5
5	4
6	6
7	5
n=8	3

2. version: guard-based



Problem:

There are m guards in the City Watch, and we know which guard was on duty on each of the n days. Which guard was on duty **most** days?

day	guard		guards=	1	2	3	4	5	6	7
1	5							x		
2	3					x				
3	4						x			
4	5							x		
5	4						x			
6	6								x	
7	3					x				
8	5							x		
			cnt=	0	0	2	2	3	1	0

2. version: guard-based



Problem:

There are m guards in the City Watch, and we know which guard was on duty on each of the n days. Which guard was on duty **most days**?

Idea:

1. For each **guard**, we determine **how many times** he had duty
 2. We choose the **guard** where **this number** is **the highest**
- => **counting** in a **maximum selection**

guards=	1	2	3	4	5	6	7
					x		
			x				
				x			
					x		
				x			
						x	
			x				
					x		
cnt=	0	0	2	2	3	1	0

2. version: guard-based



Problem:

There are m guards in the City Watch, and we know which guard was on duty on each of the n days. Which guard was on duty **most days**?

Specification:

Íŋ ŋ ∈ ℕ ŋ ∈ ℕ ġũăđ ∈ ℕ , ŋ

Ôűť đűťŷ ∈ ℕ

Gŋ ħŋ ℕ ℕ ħŋ xġ CÔŮŦŦ î , ŋ ġũăđ î xġ

Rsê

Rộšť đűťŷ ŃĂŸ xġ , ŋ ħŋ xġ

guards=	1	2	3	4	5	6	7
					x		
			x				
				x			
					x		
				x			
						x	
			x				
					x		
cnt=	0	0	2	2	3	1	0



2. version: guard-based

Algorithm:



guards=	1	2	3	4	5	6	7
					x		
			x				
				x			
					x		
				x			
						x	
			x				
					x		
cnt=	0	0	2	2	3	1	0



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PROGRAMMING – Zsuzsa Pluhár

Construction with copy - datatransformation

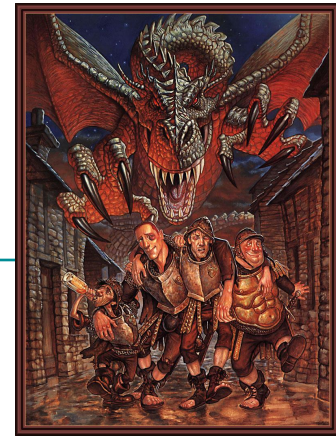
Problem:

There are m guards in the City Watch, and we know for each guard from which day to which day he was on duty. When was the city (Ankh-Morpork City) most protected?



Illustration by Paul Kidby (from Terry Pratchett's Discworld series)

Construction with copy - datatransformation

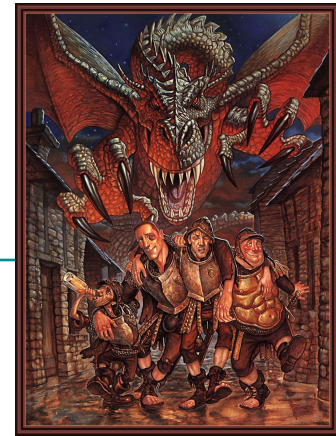


Problem:

There are m guards in the City Watch, and we know for each guard from which day to which day he was on duty. When was the city (Ankh-Morpork City) most protected?

guard	from	to	days=	1	2	3	4	5	6	7	8	9	10
1	1	3		x	x	x							
2	2	2			x								
3	5	8						x	x	x	x		
4	7	8								x	x		
5	8	9									x	x	
			cnt=	1	2	1	0	1	1	2	3	1	0

day based control



Problem: ... When was the city most protected?

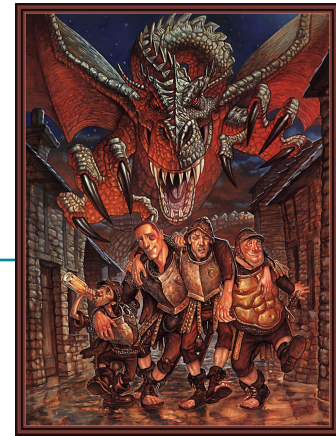
Idea:

1. **Transformation:** For **each day**, we determine **how many guards** were on duty at that time
2. Which day has the **highest number** of guards?

=> **counting** in a **copy** and **maximum selection**

guard	from	to		days=	1	2	3	4	5	6	7	8	9	10
1	1	3			x	x	x							
2	2	2				x								
3	5	8							x	x	x	x		
4	7	8									x	x		
5	8	9										x	x	
				cnt=	1	2	1	0	1	1	2	3	1	0

1. version: every day...



Problem: ... When was the city most protected?

Specification:

Ín $n \in \mathbb{N}$ $n \in \mathbb{N}$ $g_{u\grave{a}s\grave{d}s} \in \mathbb{T} \text{ in } \mathbb{T} \text{ es } \acute{w}\acute{a}\acute{l}\acute{l}$, n
 $\mathbb{T} \text{ in } \mathbb{T} \text{ es } \acute{w}\acute{a}\acute{l}\acute{l}$ $\acute{g}s\acute{o}n$ N y $\mathbb{T} \acute{o}$ N

Auy. $\mathbb{T} \text{ in } \mathbb{T} \in N$, n

Ôut $n\grave{a}y \text{ in } \mathbb{T} \in N$

Gn $n\grave{g}u\grave{a}s\grave{d}s$ N N

$n\grave{g}u\grave{a}s\grave{d}s$ $\acute{d}\acute{a}y$ $C\acute{O}U\mathbb{T}$ $\acute{i}$, n $g_{u\grave{a}s\grave{d}s}$ $\acute{i}$ $\acute{g}s\acute{o}n$ $\acute{d}\acute{a}y$ $g_{u\grave{a}s\grave{d}s}$ $\acute{i}$ $\mathbb{T} \acute{o}$

$R\acute{s}\acute{e}$ n . $\acute{a}\acute{n}\acute{d}$ $\forall i \in$, n $g_{u\grave{a}s\grave{d}s}$ $\acute{i}$ $\acute{g}s\acute{o}n$, $\acute{a}\acute{n}\acute{d}$ $g_{u\grave{a}s\grave{d}s}$ $\acute{i}$ $\mathbb{T} \acute{o}$ n

$R\acute{o}\mathbb{T}$ $\forall \acute{d}\acute{a}y \in$, n $\mathbb{T} \text{ in } \mathbb{T}$ $\acute{d}\acute{a}y$ $n\grave{g}u\grave{a}s\grave{d}s$ $\acute{d}\acute{a}y$ $\acute{a}\acute{n}\acute{d}$
 $n\grave{a}y \text{ in } \mathbb{T}$ $\acute{N}\acute{A}\acute{Y}$ $\acute{d}\acute{a}y$, n $\mathbb{T} \text{ in } \mathbb{T}$ $\acute{d}\acute{a}y$

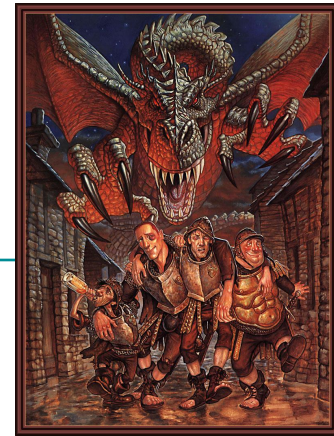
guard	from	to	days=	1	2	3	4	5	6	7	8	9	10
1	1	3		x	x	x							
2	2	2			x								
3	5	8						x	x	x	x		
4	7	8								x	x		
5	8	9									x	x	
cnt=				1	2	1	0	1	1	2	3	1	0



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2. version: copy



Problem: ... When was the city most protected?

Specification:

Ín $n \in \mathbb{N}$ $n \in \mathbb{N}$ $g_{u\grave{a}s\grave{d}s} \in \mathbb{T} \text{ in } \mathbb{T} \text{ es } \acute{w}\acute{a}\acute{l}\acute{l}$, n
 $\mathbb{T} \text{ in } \mathbb{T} \text{ es } \acute{w}\acute{a}\acute{l}\acute{l}$ $\acute{g}\acute{s}\acute{o}\acute{n}$ $\mathbb{N}$ y $\mathbb{T}$ $\acute{o}$ $\mathbb{N}$

Auy. $\mathbb{C} \text{ in } \mathbb{T} \in \mathbb{N}$, n

Ôut $n\grave{a}y \text{ in } \mathbb{D} \in \mathbb{N}$

Gn $h\grave{n}g\grave{u}\acute{a}s\grave{d}s$ $\mathbb{N}$ $\mathbb{N}$

$h\grave{n}g\grave{u}\acute{a}s\grave{d}s$ $\acute{d}\acute{a}y$ $\mathbb{C}\mathbb{O}\mathbb{U}\mathbb{N}\mathbb{T}$ $\acute{i}$, n $g\grave{u}\acute{a}s\grave{d}s$ $\acute{i}$ $\acute{g}\acute{s}\acute{o}\acute{n}$ $\acute{d}\acute{a}y$ $g\grave{u}\acute{a}s\grave{d}s$ $\acute{i}$ $\mathbb{T}$ $\acute{o}$

R $\acute{s}\acute{e}$ n . $\acute{a}\acute{n}\acute{d}$ $\forall i \in$, n $g\grave{u}\acute{a}s\grave{d}s$ $\acute{i}$ $\acute{g}\acute{s}\acute{o}\acute{n}$, $\acute{a}\acute{n}\acute{d}$ $g\grave{u}\acute{a}s\grave{d}s$ $\acute{i}$ $\mathbb{T}$ $\acute{o}$ n

R $\acute{o}\mathbb{T}$ $\mathbb{C} \text{ in } \mathbb{T}$ $\mathbb{C}\mathbb{O}\mathbb{R}\mathbb{Y}$ $\acute{d}\acute{a}y$, n $h\grave{n}g\grave{u}\acute{a}s\grave{d}s$ $\acute{d}\acute{a}y$ $\acute{a}\acute{n}\acute{d}$
 $n\grave{a}y \text{ in } \mathbb{D}$ $\mathbb{N}\mathbb{A}\mathbb{Y}$ $\acute{d}\acute{a}y$, n $\mathbb{C} \text{ in } \mathbb{T}$ $\acute{d}\acute{a}y$

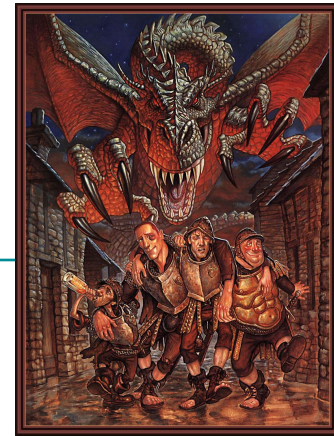
guard	from	to		days=	1	2	3	4	5	6	7	8	9	10
1	1	3			x	x	x							
2	2	2				x								
3	5	8							x	x	x	x		
4	7	8									x	x		
5	8	9										x	x	
cnt=					1	2	1	0	1	1	2	3	1	0



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2. version: copy



Problem: ... When was the city most protected?

Specification:

Íñ ñ ∈ Ñ ñ ∈ Ñ ġũấđđ ∈ Ĳĩñđêšwấỉỉ , ñ
Ĳĩñđêšwấỉỉ ģsộñ Ñ ỵ. ặ ñ

Auy. ặ ñ ∈ Ñ , ñ

Ôủđ ñấỵĩñđ ∈ Ñ

Gñ ģñũấđđ Ñ Ñ

ģñũấđđ đấỵ CÔỦỦ ĩ , ñ ģũấđđ ĩ ģsộñ đấỵ ģũấđđ ĩ ặ

Rsê ñ . ấđđ ∀ ĩ ∈ , ñ ģũấđđ ĩ ģsộñ , ấđđ ģũấđđ ĩ ặ ñ

Rộđđ ặ ñ đấỵ CÔỠỠ đấỵ , ñ ģñũấđđ đấỵ ấđđ
ñấỵĩñđ ÑÃỠ đấỵ , ñ ặ đấỵ

guard	from	to	days=	1	2	3	4	5	6	7	8	9	10
1	1	2											
2	2	3			x								
3	5	8						x	x	x	x		
4	7	8								x	x		
5	8	9									x	x	
cnt=				1	2	1	0	1	1	2	3	1	0

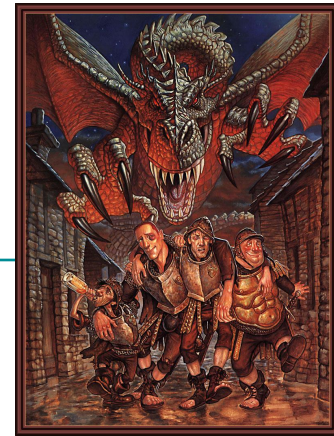
COPY can be combined with the next used pattern



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3. version: combining copy



Problem: ... When was the city most protected?

Specification:

Ín $n \in \mathbb{N}$ $n \in \mathbb{N}$ $g_{u\grave{a}sd\grave{s}} \in \mathbb{T} \text{ in } t\grave{e}s\grave{w}\grave{a}l\grave{l}$, n
 $\mathbb{T} \text{ in } t\grave{e}s\grave{w}\grave{a}l\grave{l}$ $g\grave{s}o\grave{n}$ $\mathbb{N}$ y $t\grave{o}$ $\mathbb{N}$

~~Auy $\mathbb{C} \text{ in } t\grave{e} \in \mathbb{N}$, n~~

Ôut $n\grave{a}y \text{ in } d \in \mathbb{N}$

Gn $h\grave{n}g\grave{u}u\grave{a}sd\grave{s}$ $\mathbb{N}$ $\mathbb{N}$

$h\grave{n}g\grave{u}u\grave{a}sd\grave{s}$ $d\grave{a}y$ $C\grave{O}U\grave{N}T$ $\hat{i}$, n $g\grave{u}u\grave{a}sd\grave{s}$ $\hat{i}$ $g\grave{s}o\grave{n}$ $d\grave{a}y$ $g\grave{u}u\grave{a}sd\grave{s}$ $\hat{i}$ $t\grave{o}$

R $\hat{s}\hat{e}$ n . $\grave{a}n\grave{d}$ $\forall i \in$, n $g\grave{u}u\grave{a}sd\grave{s}$ $\hat{i}$ $g\grave{s}o\grave{n}$, $\grave{a}n\grave{d}$ $g\grave{u}u\grave{a}sd\grave{s}$ $\hat{i}$ $t\grave{o}$ n

R $\hat{o}t\grave{t}$ $n\grave{a}y \text{ in } d$ $\grave{N}\hat{A}\hat{Y}$ $d\grave{a}y$, n $h\grave{n}g\grave{u}u\grave{a}sd\grave{s}$ $d\grave{a}y$

guard	from	to		days=	1	2	3	4	5	6	7	8	9	10
1	1	3			x	x	x							
2	2	2				x								
3	5	8							x	x	x	x		
4	7	8									x	x		
5	8	9										x	x	
				cnt=	1	2	1	0	1	1	2	3	1	0



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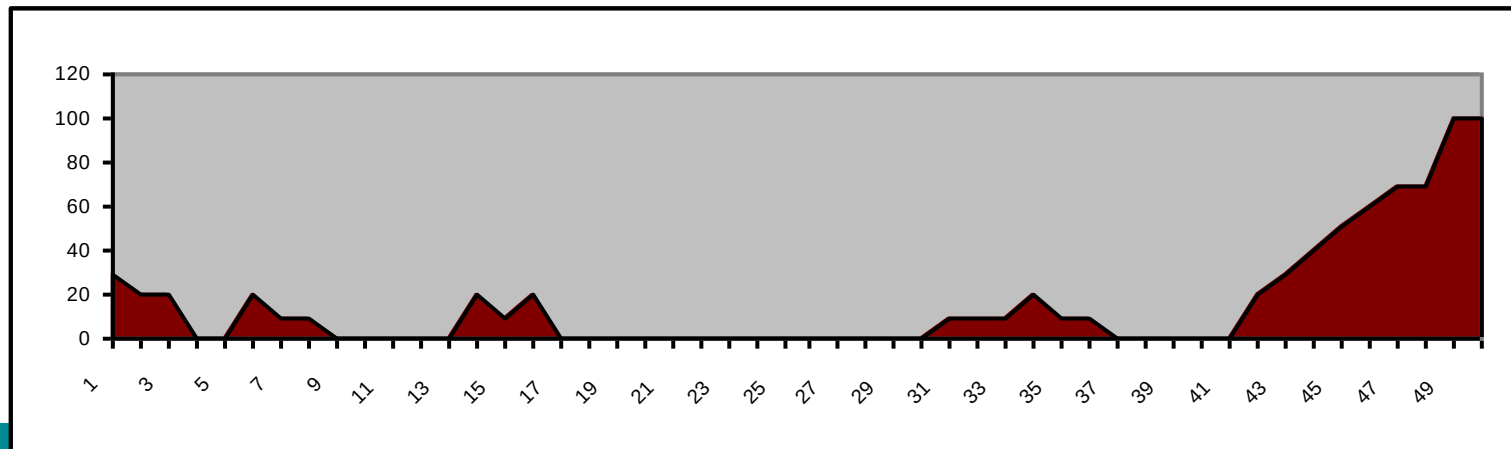
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Problems with/on interval

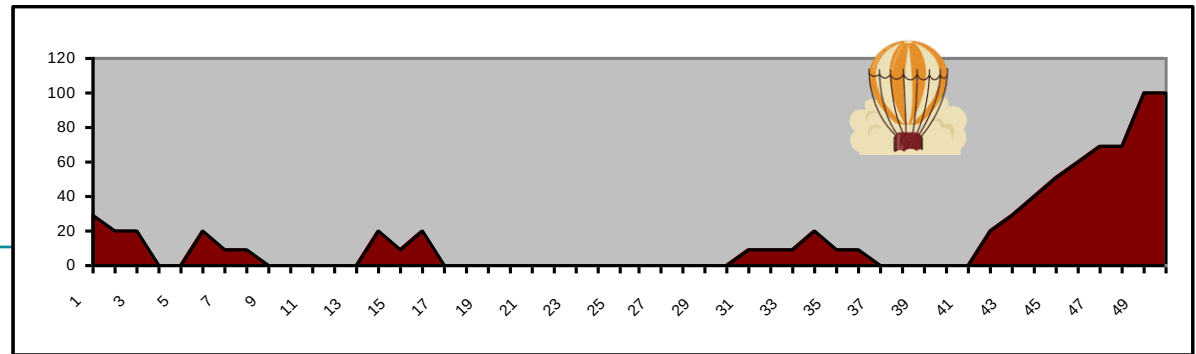


Problem:

Twoflower, the first tourist ever on the Discworld, measured the height every kilometers (≥ 0) during a balloon flight. He measured 0 where there is sea and >0 where there is land. Give us the widest island!



1. version



Idea:

1. Let's choose the island beginnings that **have an end**!
2. **Copy** the island ends to them (can be merged)
3. Find the distances by **copying** (can be merged)
4. Determine the **maximum distance**, if any!

đêğîñê îşlăñđênđ îşlăñđşţăşt
ŃÍŞ ŞÉARÇĤ êyîşţş DÉCÍŞÍÔŇ
CÔRŸ ŞÉARÇĤ îñđ ŞÉLÉCŢÍÔŇ
çăłçułăţîîñ CÔRŸ
ŃĂŸ ÍĞŃĂŸ



1. version

Specification:

În $\eta \in \mathbb{N}$ $\text{h\^e i\^g h\^t} \in \mathbb{N}$, η

Au η $\text{c}\eta\text{t} \in \mathbb{N}$ $\text{c}\eta\text{g i}\eta\text{s} \in \mathbb{N}$, $\text{c}\eta\text{t}$ $\text{n\^a y i}\eta\text{d} \in \mathbb{N}$

Ôu t $\text{c} \in \mathbb{N}$ $\text{e} \in \mathbb{N}$

G η $\text{i}\text{s l\^a n d s t j \^a s t}$ $\mathbb{N}$ $\mathbb{L}$

$\text{i}\text{s l\^a n d s t j \^a s t}$ i $\text{g\^a l s\^e i}\eta\text{ i}$,
 $\text{h\^e i\^g h\^t i}$, . $\text{\^a n d h\^e i\^g h\^t i}$. $\text{\^o t j h\^e s x i s\^e}$

G η $\text{i}\text{s l\^a n d\^e n d}$ $\mathbb{N}$ $\mathbb{L}$

$\text{i}\text{s l\^a n d\^e n d}$ i $\text{g\^a l s\^e i}\eta\text{ i}$ η
 $\text{h\^e i\^g h\^t i}$, . $\text{\^a n d h\^e i\^g h\^t i}$. $\text{\^o t j h\^e s x i s\^e}$

G η $\text{s\^e \^a s c h\^e n d}$ $\mathbb{N}$ $\mathbb{L}$ $\mathbb{Y}$ $\mathbb{N}$

$\text{s\^e \^a s c h\^e n d i}$ $\text{\^S\^E\^A\^R\^C\^H k i}$ η $\text{i}\text{s l\^a n d\^e n d k}$

G η d i s t j $\mathbb{N}$ $\mathbb{N}$ d i s t j i $\text{s\^e \^a s c h\^e n d i}$ $\text{i}\eta\text{d i}$

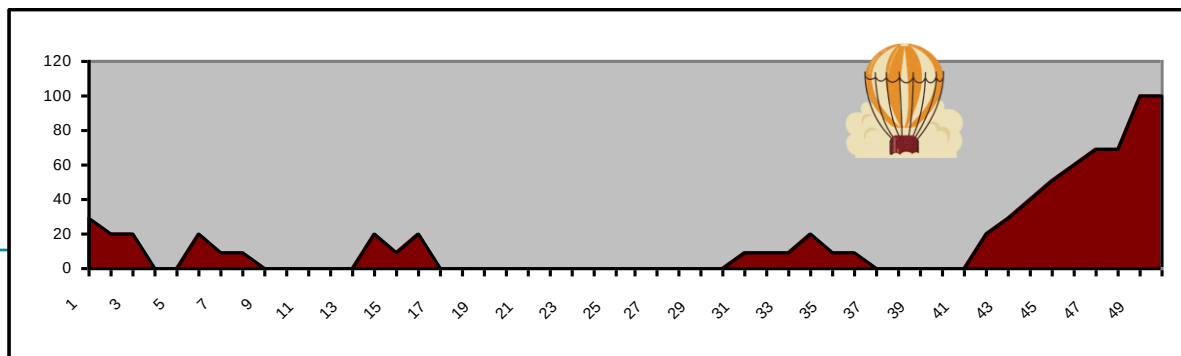
R $\text{s\^e}$

Rô st j $\text{c}\eta\text{t}$ $\text{c}\eta\text{g i}\eta\text{s}$ N i s i , η $\text{i}\text{s l\^a n d s t j \^a s t i}$ $\text{\^a n d s\^e \^a s c h\^e n d i}$ e y i s t j s i $\text{\^a n d}$

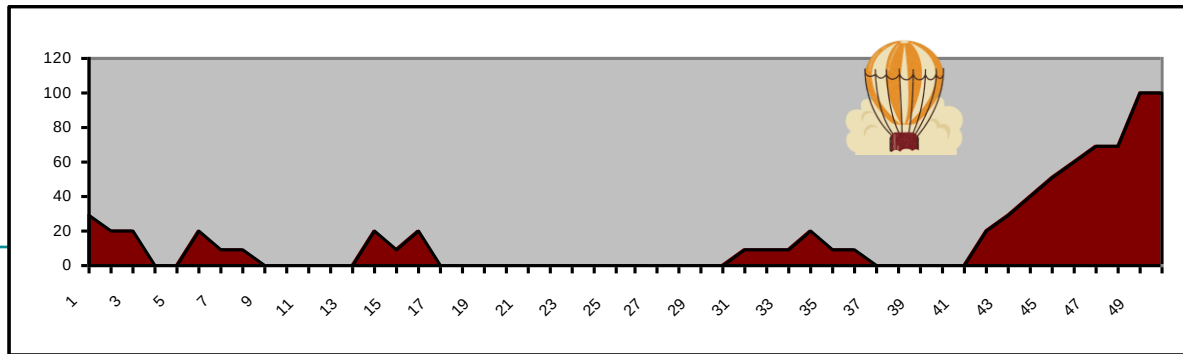
$\text{c}\eta\text{t}$. $\text{n\^a y i}\eta\text{d}$ $\text{N\^A Y i}$, $\text{c}\eta\text{t}$ $\text{d i s t j c}\eta\text{g i}\eta\text{s i}$ $\text{\^a n d}$

$\text{c}\eta\text{g i}\eta\text{s}$ $\text{n\^a y i}\eta\text{d}$ $\text{\^a n d e}$ $\text{s\^e \^a s c h\^e n d c}$ $\text{i}\eta\text{d}$ $\text{\^a n d}$

$\text{c}\eta\text{t}$. c . $\text{\^a n d e}$.



1. version



Specification:

$$\exists \eta \in \mathbb{N} \quad \forall t \in \mathbb{N}, \quad \eta \leq t$$

Auy, $\varphi_{\eta f} \in N$ $\check{c}\hat{e}\hat{g}\hat{i}\eta s \in N$, $\varphi_{\eta f}$ $n\grave{a}y.\hat{i}\eta d \in N$

$$\hat{O}_{\text{uf}} \quad \check{c} \in \mathbb{N} \quad \hat{e} \in \mathbb{N}$$

Gn ỉs'ẩnđs'ẩs'ẩ N L'

îşlăndştǎştǎ î ǵǎlşê îğ î ,

hệ thống i , ă ã đ hệ thống i . ộ thê s x i s e

Gn i s l ấ n đ ê n đ N L '

îşlândênd î gǎlşê îğ î ȳ

height in , and height in . Otherwise

Gn ợăăợhệđ N L y N

şêăşçêñđ î ŞÉARCH k î ȧ îşlăñđêñđ k

Gn đĩstj N N đĩstj i şeăşçhẽđ i iñđ i

Rsê

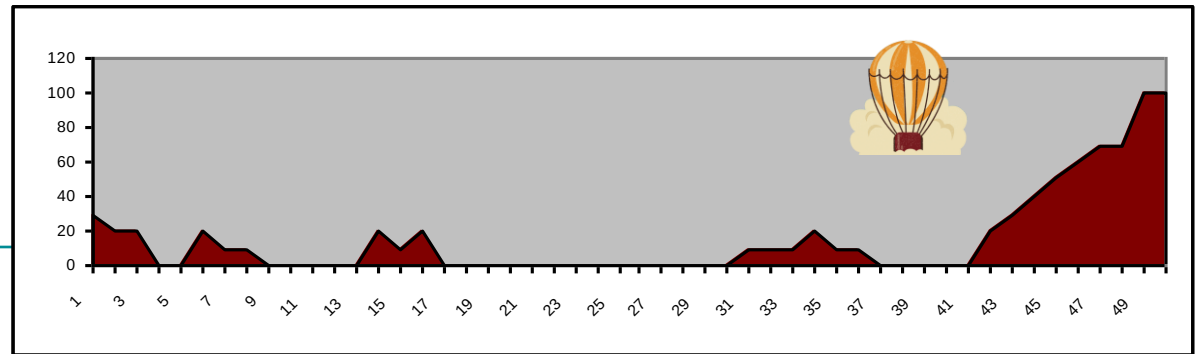
Rôstf çntf čêgîns ĆÍȘ î ȧ îsl'ăndstfăstf î ănd sêăschênd î êyîstfs î ănd

chữ này inđ ÑÂ ì chữ đistf cêgîns ì ănd

č čēgīns náyīnd ānd ē sēāsčhēnd č īnd ānd

çŋtʃ č ăŋd ê .

2. version



with conditional maximum selection

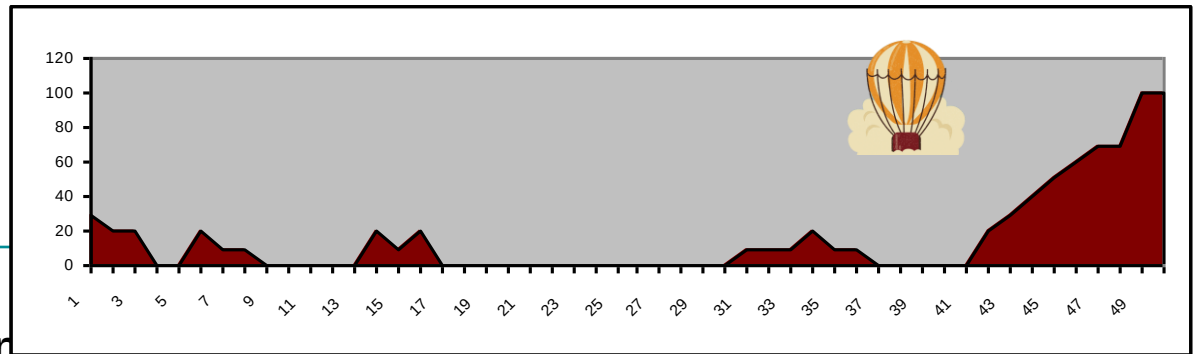
Specification:

Rõşť çηť čêġîņş ŃÍŞ î , η îşĺăηđşťăşť î ■ î ăηđ

êy.îşťş ăăy.îηđ

CÔNDŃAY î , çηť đîşť čêġîņş î şêăşçêηđ î êy.îşťş ăηđ
êy.îşťş č čêġîņş ăăy.îηđ ăηđ ê şêăşçêηđ č îηđ

3. version



Specification:

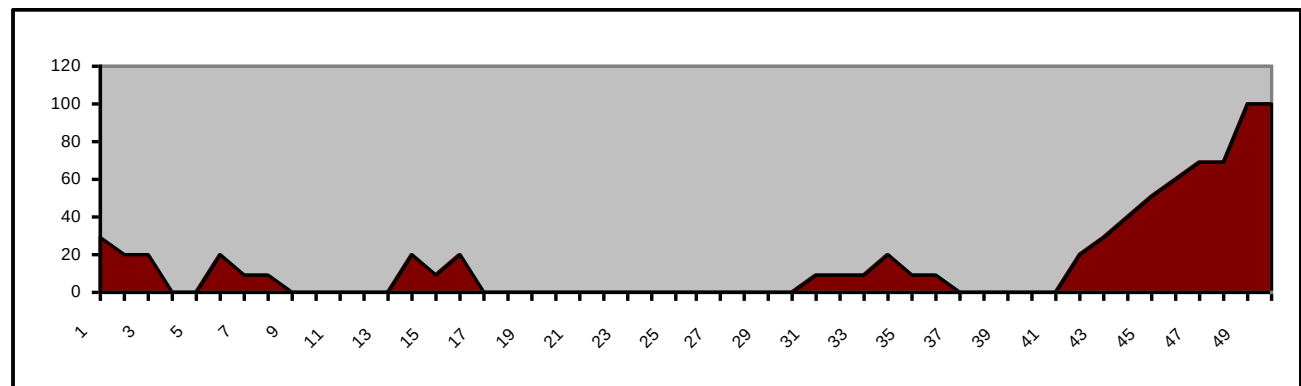
$\text{In } n \in \mathbb{N} \text{ h\u00e9i\u00f1htj} \in \mathbb{N}, n$
 $\text{Outj } \text{eyi\u00f1tj} \in L, \text{ c} \in \mathbb{N}, \text{ e} \in \mathbb{N}$
 $\text{Gn } \text{isl\u00e1ndstj\u00e1stj } N, L$
 $\text{isl\u00e1ndstart } i \text{ g\u00e1ls\u00e9 } i \text{ g} i$
 $\text{h\u00e9i\u00f1htj } i, \text{ . \u00e1nd h\u00e9i\u00f1htj } i \text{ . \u00f1th\u00e9sx\u00e9}$
 $\text{Gn } \text{isl\u00e1nd\u00e9nd } N, L$
 $\text{isl\u00e1nd\u00e9nd } i \text{ g\u00e1ls\u00e9 } i \text{ g} i n$
 $\text{h\u00e9i\u00f1htj } i, \text{ . \u00e1nd h\u00e9i\u00f1htj } i \text{ . \u00f1th\u00e9sx\u00e9}$
 $\text{Gn } \text{s\u00e9\u00e1s\u00e7h\u00e9nd } N, L, y, N$
 $\text{s\u00e9\u00e1s\u00e7h\u00e9nd } i \text{ S\u00c9ARCH } k, i, n \text{ isl\u00e1nd\u00e9nd } k$
 $\text{Gn } \text{d\u00edstj } N, N \text{ dist } i \text{ s\u00e9\u00e1s\u00e7h\u00e9nd } i \text{ \u00ednd } i$
 $\text{Rs\u00e9}$
 $\text{R\u00f1stj } \text{eyi\u00f1tj} \text{ c}$
 $\text{COND\u00d1AY } i, n \text{ d\u00edstj } i \text{ isl\u00e1ndstj\u00e1stj } i \text{ \u00e1nd s\u00e9\u00e1s\u00e7h\u00e9nd } i \text{ eyi\u00f1tj}$
 $\text{\u00e1nd}$
 $\text{eyi\u00f1tj} \text{ \u00e9 s\u00e9\u00e1s\u00e7h\u00e9nd } \text{c} \text{ \u00ednd}$

Problems with/on interval



Verisons:

- the continents are also islands (at the rands, end, begin)
- we search for the shortest island
- we search for the longest section of the see without island
- ...



Interval again

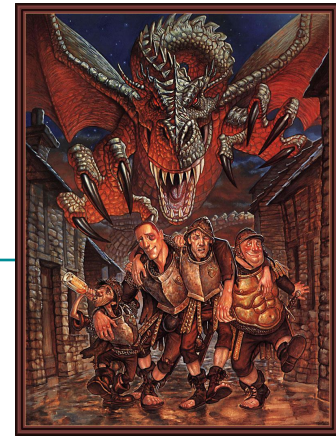
Problem:

There are m guards in the City Watch, and we know for each guard from which day to which day he was on duty. What was the **longest period** during which the city was protected?



Illustration by Paul Kidby (from Terry Pratchett's Discworld series)

Interval again

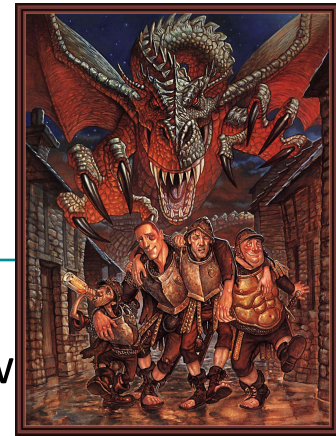


Problem:

There are m guards in the City Watch, and we know for each guard from which day to which day he was on duty. What was the **longest period** during which the city was protected?

guard	from	to	days=	1	2	3	4	5	6	7	8	9	10
1	1	3		x	x	x							
2	2	2			x								
3	5	8						x	x	x	x		
4	7	8								x	x		
5	8	9									x	x	
			cnt=	1	2	1	0	1	1	2	3	1	0

1. version



Problem: ... What was the longest period during which the city was protected?

Specification:

Ín $n \in \mathbb{N}$ $n \in \mathbb{N}$ $g_{u\grave{a}s\grave{d}\grave{s}} \in \mathbb{T} \cap \mathbb{T} \in \mathbb{S} \hat{w} \hat{a} \hat{l} \hat{l}$, n
 $\mathbb{T} \cap \mathbb{T} \in \mathbb{S} \hat{w} \hat{a} \hat{l} \hat{l}$ $\grave{s} \hat{s} \hat{o} \hat{n}$ N y $\mathbb{T} \hat{o}$ N

Auy $\mathbb{C} \cap \mathbb{T} \in \mathbb{N}$, n

Ôu $\mathbb{T}$ $n \grave{a} y \mathbb{i} \cap \mathbb{D} \in \mathbb{N}$

G $\mathbb{n}$ $\mathbb{h} \cap \mathbb{g} \cap \mathbb{u} \cap \mathbb{a} \cap \mathbb{s} \cap \mathbb{d} \cap \mathbb{s} \cap \mathbb{N}$ $\mathbb{N}$

$\mathbb{h} \cap \mathbb{g} \cap \mathbb{u} \cap \mathbb{a} \cap \mathbb{s} \cap \mathbb{d} \cap \mathbb{s} \cap \mathbb{d} \grave{a} y$ $\mathbb{C} \hat{O} \hat{U} \mathbb{N} \mathbb{T}$ $\mathbb{i}$, n $\mathbb{g} \cap \mathbb{u} \cap \mathbb{a} \cap \mathbb{s} \cap \mathbb{d} \cap \mathbb{s} \cap \mathbb{i}$ $\grave{s} \hat{s} \hat{o} \hat{n}$ $\mathbb{d} \grave{a} y$ $\mathbb{g} \cap \mathbb{u} \cap \mathbb{a} \cap \mathbb{s} \cap \mathbb{d} \cap \mathbb{s} \cap \mathbb{i}$ $\mathbb{T} \hat{o}$

R $\mathbb{s} \hat{e}$ n . $\grave{a} \mathbb{n} \mathbb{d}$ $\forall i \in$, n $\mathbb{g} \cap \mathbb{u} \cap \mathbb{a} \cap \mathbb{s} \cap \mathbb{d} \cap \mathbb{s} \cap \mathbb{i}$ $\grave{s} \hat{s} \hat{o} \hat{n}$, $\grave{a} \mathbb{n} \mathbb{d}$ $\mathbb{g} \cap \mathbb{u} \cap \mathbb{a} \cap \mathbb{s} \cap \mathbb{d} \cap \mathbb{s} \cap \mathbb{i}$ $\mathbb{T} \hat{o}$ n

R $\hat{o} \mathbb{s} \mathbb{T}$ $\mathbb{C} \cap \mathbb{T}$ $\mathbb{C} \hat{O} \mathbb{R} \mathbb{Y}$ $\mathbb{d} \grave{a} y$, n $\mathbb{h} \cap \mathbb{g} \cap \mathbb{u} \cap \mathbb{a} \cap \mathbb{s} \cap \mathbb{d} \cap \mathbb{s} \cap \mathbb{d} \grave{a} y$ $\grave{a} \mathbb{n} \mathbb{d}$

let's continue with the solution of the „interval problems” on cnt[]

guard	from	to	days=	1	2	3	4	5	6	7	8	9	10
1	1	3		x	x	x							
2	2	2			x								
3	5	8						x	x	x	x		
4	7	8								x	x		
5	8	9									x	x	
cnt=				1	2	1	0	1	1	2	3	1	0

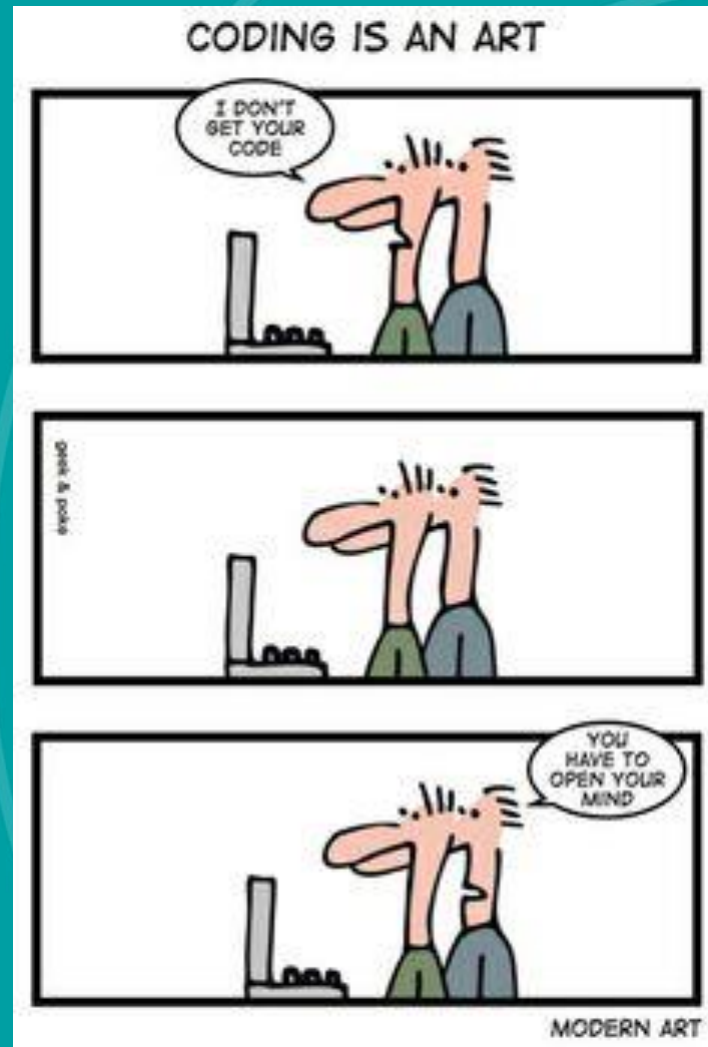


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Thank you for your attention!